

GRIDNEST ENERGY

GridNest W11

Home wallbox - installation and operating data

Model	W11 Home
Product code	GN-W11-01
Rated output	11 kW
Edition	Installation guide 2.1 - 2026

Technical information for a qualified installer and the owner of the installed GridNest W11 home charging unit.

Read the identification, safety and applicability information before relying on this document.

Contents

Section	Page
Product overview	2
Electrical ratings	3
Protection	4
Installation	5
Solar-surplus mode	6
Load management	7
Communications	8
Status and faults	9
Technical summary	10

3. Electrical ratings

Rated charging output

The W11 supplies up to 11 kW on a three-phase 400 V circuit at 16 A per phase.

Current range

The charging current is adjustable from 6 A to 16 A per phase. Dynamic control may reduce current during household load peaks or low solar surplus.

Vehicle limit

Delivered power cannot exceed the vehicle onboard-charger limit. A vehicle rated at 11 kW AC remains limited to 11 kW.

Supply	3-phase 400 V AC, 50 Hz
Maximum current	16 A per phase
Maximum output	11 kW
Vehicle connector	Type 2, 5 m tethered cable
Efficiency	>98% power-path efficiency
Ingress rating	IP54

4. Protective functions

Residual current protection

The W11 contains 6 mA DC residual-current detection. The upstream installation requires the RCD and overcurrent protection stated in the site design.

Protective earth

Charging is inhibited when the protective-earth check fails.

Temperature

The unit reduces current or stops if internal temperature exceeds its safe range.

No direct DC input

The W11 accepts AC supply only. Photovoltaic strings must not be connected to the unit.

5. Installation requirements

Dedicated circuit

Install on a dedicated three-phase circuit sized for continuous 16 A per phase operation.

Isolation

Provide an accessible isolator and label the distribution board circuit.

Commissioning

Record model, serial, protective-device test results, phase voltage and configured maximum current.

Qualified installer

Installation and test must be performed by a person qualified for the applicable electrical system.

6. Solar-surplus charging

Energy-manager input

The W11 can receive a dynamic current setpoint over Modbus TCP from a compatible energy manager or inverter gateway.

Operating range

Solar-surplus mode varies charging from approximately 1.4 kW single-phase equivalent control up to the configured 11 kW maximum, subject to site configuration.

HelioSync integration

Protocol profile HS-EV-2 is compatible with the HelioSync I10 gateway when both devices use firmware listed in their current guides.

Grid supplement

When the site setting permits grid import, the grid supplies any difference between photovoltaic output, household demand and the requested charging power. The inverter's PV output alone does not necessarily cap wallbox output.

Solar-surplus charging still uses the site's AC distribution system, protective devices and wallbox. It is not a direct panel-to-vehicle connection.

7. Dynamic load management

Site limit

A current transformer or smart meter measures the building connection. The controller reduces wallbox current before the configured import limit is exceeded.

Phase balance

Three-phase charging requires valid measurements on all phases.

Priority

Household essential loads take priority unless the installer configures a different policy.

Minimum current

If the available setpoint falls below the permitted minimum, the session pauses and resumes when capacity returns.

8. Communications

Ethernet

100BASE-T Ethernet is the preferred commissioning connection.

Wi-Fi

2.4 GHz Wi-Fi is available where signal quality is adequate.

Modbus TCP

Register profile GN-W11-2 supports current setpoint, session power, energy and state.

Offline behaviour

If communication is lost, the configured fail-safe current applies. The value must not exceed the site circuit rating.

9. Status and fault handling

Ready

Supply and protection checks are valid; no vehicle session is active.

Charging

A valid control pilot and current setpoint are present.

Paused by energy manager

The requested current is below the charging threshold; this is not necessarily a fault.

Protection fault

Isolate the unit and arrange inspection if earth, residual-current or overtemperature protection repeatedly operates.

Indicator	Meaning
Blue steady	Ready
Blue pulse	Charging
Amber	Paused or limited
Red	Fault - inspect before reset

10. Technical summary

Installed identity

Use the site connection schedule to confirm that GN-W11-01 is the installed unit.

Source control

Source ID s6-wallbox-datasheet; installation guide 2.1.

Item	GridNest W11 Home
Product code	GN-W11-01
Rated output	11 kW
Supply	3-phase 400 V, 16 A/phase
Current setting	6-16 A per phase
Connector	Type 2
Solar integration	Modbus TCP profile HS-EV-2
Direct PV DC input	Not supported